SYM-H DLM Modelling with Bz-GSM

Introduction and motivation

Distributed lag models (DLMs) are statistical models designed to estimate relationships in which the effect of a predictor is distributed over time rather than occurring entirely at the instant at which the predictor is observed. Although DLMs were originally developed in time-series and econometric setings, they have subsequently been widely applied in environmental and epidemiological research such as Dempsey and Wyse (2025) and Gasparrini et al. (2011).

The same modelling idea is potentially useful in heliophysics, where changes in the solar wind can influence the terrestrial magnetosphere over a range of time scales. Of particular importance is the north–south component of the interplanetary magnetic field, Bz. When Bz is southward, magnetic reconnection between the interplanetary and terrestrial magnetic fields is enhanced, allowing more efficient transfer of solar-wind energy into the magnetosphere. The subsequent geomagnetic response is therefore not necessarily instantaneous and may depend on the preceding history of the solar-wind conditions. Burton, McPherron and Russell (1975), for example, developed an empirical model for Dst based on the solar-wind velocity, density and north–south interplanetary magnetic-field component, incorporating both energy injection and subsequent ring-current decay.

Geomagnetic storm intensity is commonly described using the disturbance storm time (Dst) index, measured in nanotesla (nT). In this study, however, the response variable is the SYM-H index, which provides a higher-temporal-resolution measure of the symmetric disturbance in the horizontal geomagnetic field. Whereas Dst is traditionally reported at hourly resolution, SYM-H is available at minute resolution and therefore captures short-timescale storm variations and minima more precisely. Zesta and Oliveira (2019) consequently used SYM-H rather than Dst when classifying storm intensity.

The storm categories used in this study are:

Minimum SYM-H	**Interpretation**
> -50 nT	Weak
$-100 < \mathrm{SYM-H} \leq -50$ nT	Moderate
$-150 < \mathrm{SYM-H} \leq -100$ nT	Strong
$-250 < \mathrm{SYM-H} \leq -150$ nT	Severe
< -250 nT	Extreme

These thresholds follow the classification used by Zesta and Oliveira; the authors note that the intensity intervals are an empirical classification rather than universal physical boundaries.

The solar-wind and geomagnetic data used here were obtained from NASA's OMNIWeb High Resolution OMNI database. High-resolution OMNI provides 1-minute and 5-minute solar-wind magnetic-field and plasma measurements shifted to the Earth's bow-shock nose, together with geomagnetic activity indices including SYM-H. In this analysis, the 5-minute data were used.

A basic distributed lag model can be written as

$$Y_t = \alpha + \sum_{\ell=0}^{L} \beta_\ell X_{t-\ell} + \varepsilon_t$$

where Y_t is the response variable, α is the intercept, $X_{t-\ell}$ represents the predictor observed ℓ time steps previously, and β_ℓ represents the contribution associated with that lag. Thus, β_0 describes the contemporaneous association, while $\beta_1, \beta_2, \ldots, \beta_L$ describe progressively earlier predictor values.

In this application, X initially represents IMF Bz, while Y represents SYM-H. Because the data are recorded every five minutes, estimating a separate unrestricted coefficient at every lag can produce a large number of parameters and a highly irregular estimated lag-response curve. To reduce the dimensionality of the lag structure, the lag coefficients were therefore represented using a B-spline basis. Rather than estimating everybeta_l independently, the spline model estimates a smaller number of basis coefficients from which a smooth lag-response curve is reconstructed. This provides local flexibility while constraining neighboring lag effects to vary more smoothly. This general use of basis functions to represent lag-response relationships is consistent with distributed-lag methodology.

However, SYM-H cannot realistically be expected to depend only on the preceding history of Bz. The current state of the magnetosphere contains substantial information about its near-future state. The model was therefore extended to a direct autoregressive distributed lag formulation:

$$Y_{t+h} = \alpha + \phi Y_t + \sum_{\ell=0}^{L} \beta_\ell X_{t-\ell} + \varepsilon_{t+h}$$

where Y_t is the currently observed SYM-H value and Y_{t+h} is SYM-H at a future forecast horizon h.

For the main specification, the forecast horizon was set to 120 minutes. With five-minute observations, this corresponds to

$$h = 24$$

time steps. The model therefore asks whether the current geomagnetic state together with recent solar-wind conditions contains useful information for predicting SYM-H two hours later.

A further extension incorporates solar-wind bulk speed:

$$Y_{t+h} = \alpha + \phi Y_t + \sum_{\ell=0}^{L} \beta_\ell Bz_{t-\ell} + \sum_{\ell=0}^{L} \gamma_\ell V_{t-\ell} + \varepsilon_{t+h}$$

where V denotes solar-wind speed. Including speed is physically motivated by established solar-wind coupling models. For example, Burton et al. related geomagnetic activity to the solar-wind electric field, which depends jointly on solar-wind velocity and the north-south IMF component rather than on Bz alone.

2. Data

The data used in this study were obtained from NASA's OMNIWeb High Resolution OMNI database, which provides near-Earth solar-wind magnetic-field and plasma measurements together with geomagnetic activity indices. Both 1-minute and 5-minute averaged data are available; the 5-minute resolution data were used here to retain short-timescale solar-wind variation while reducing the number of highly correlated consecutive observations used in the distributed lag models.

Three variables were downloaded for each event: SYM-H, Bz in GSM coordinates, and solar-wind flow speed. SYM-H was used as the response variable, while Bz and flow speed were used as solar-wind predictors.

Ten historical geomagnetic storms of differing severity were selected:

Storm	**Approx. Dst minimum (nT)**
1998-05-04	-205
2001-03-31	-387
2001-04-12	-271
2003-11-20	-422
2004-11-08	-374
2005-05-15	-247
2012-07-15	-139
2015-06-23	-198
2017-05-28	-146
2023-04-24	-213

For each event, a fixed six-calendar-day window was extracted: two calendar days preceding the storm-minimum date, the storm-minimum day itself, and three calendar days following it. The same extraction rule was used for every storm in order to avoid event-specific window selection. This interval was intended to include pre-storm conditions, the storm main phase, and a substantial part of the subsequent recovery period.

Missing solar-wind observations are an important issue in OMNI data. NASA's OMNIWeb documentation notes that gaps in plots correspond to periods for which data are unavailable, and the high-resolution data use designated fill values for missing measurements. Because the distributed lag models require a continuous history of Bz and solar-wind speed, storms containing prolonged gaps in either predictor were excluded from the validation sample. In practice, gaps comparable to or longer than the main 120-minute predictor-history window were treated as unsuitable for interpolation. Events with only short, isolated gaps were retained, and missing Bz and flow-speed observations in those events were filled using linear interpolation.

3. Final Model

The final modelling stage focused on autoregressive distributed lag models (ARDLs) for predicting future SYM-H. Two versions were considered: a Bz-only ARDL and an extended Bz + solar-wind-speed ARDL.

The Bz-only model can be written as

$$\mathrm{SYM\!-\!H}_{t+h} = \alpha + \phi\,\mathrm{SYM\!-\!H}_t + \sum_{\ell=0}^{L} \beta_\ell Bz_{t-\ell} + \varepsilon_{t+h}$$

where $\mathrm{SYM\!-\!H}_{t+h}$ is the SYM-H value h time steps ahead, $\mathrm{SYM\!-\!H}_t$ is the current observed SYM-H value, $Bz_{t-\ell}$ represents the current and past values of the IMF Bz component, and β_ℓ gives the effect associated with lag ℓ.

The extended model additionally includes solar-wind flow speed:

$$\mathrm{SYM\!-\!H}_{t+h} = \alpha + \phi\,\mathrm{SYM\!-\!H}_t + \sum_{\ell=0}^{L} \beta_\ell Bz_{t-\ell} + \sum_{\ell=0}^{L} \gamma_\ell V_{t-\ell} + \varepsilon_{t+h}$$

where $V_{t-\ell}$ denotes solar-wind speed at lag ℓ, and γ_ℓ represents the lagged contribution of solar-wind speed.

For the main specification used to illustrate the model, the forecast horizon was fixed at 120 minutes. Since the data are recorded every five minutes, this corresponds to

$$h = 24.$$

The model therefore uses information available at time t to predict SYM-H two hours later.

B-spline representation of the lag effects

A direct distributed lag model could estimate a separate coefficient for every lag,

$$\beta_0, \beta_1, \ldots, \beta_L,$$

but this becomes inefficient when many closely spaced lagged observations are included. Adjacent five-minute Bz and solar-wind-speed values are also strongly related, so estimating each lag coefficient independently can result in unstable and irregular lag-response curves.

To address this, the lag coefficients were represented using a B-spline basis. Instead of fitting a completely independent coefficient at every lag, the model expresses the lag-response curve as a combination of a smaller number of smooth basis functions.

Conceptually,

$$\beta_\ell = \sum_{k=1}^{K} \theta_k B_k(\ell),$$

where $B_k(\ell)$ are the B-spline basis functions evaluated at lag ℓ, and θ_k are the estimated basis coefficients. The same approach is used for the solar-wind-speed lag effects,

$$\gamma_\ell = \sum_{k=1}^{K} \delta_k B_k(\ell).$$

This reduces the number of parameters that must be estimated while still allowing the effect of Bz or solar-wind speed to vary flexibly over the lag window. In practical terms, nearby lag effects are encouraged to follow a smoother pattern rather than changing sharply from one five-minute interval to the next.

The Bz and speed variables use the same B-spline basis over the lag dimension, but they receive separate fitted coefficients, allowing the two predictors to have different lag-response relationships.

In the fitted models, the lag basis used five degrees of freedom (`bs_df = 5`).

Lag-window specification

The length of the predictor history is controlled in the R code through the parameter

```
max_lag <- 24
```

With five-minute observations, max_lag = 24 includes

$$Bz_t, Bz_{t-1}, \ldots, Bz_{t-24},$$

and the equivalent set of solar-wind-speed values.

Because lag 0 is included, this corresponds to 25 predictor values per variable, covering the current observation and the previous 120 minutes:

$$24 \times 5 = 120 \text{ minutes.}$$

The parameter can easily be changed to examine alternative lag windows. For example:

```
max_lag <- 12
```

uses 60 minutes of predictor history, while

```
max_lag <- 36
```

uses 180 minutes.

This provides a straightforward way to investigate whether shorter or longer histories of the solar-wind conditions improve forecast performance.

It is important to distinguish the lag window from the forecast horizon. In the main specification, both happen to equal 24 five-minute steps, but they serve different purposes. The lag window determines how much past Bz and solar-wind-speed information is supplied to the model, whereas the forecast horizon determines how far into the future SYM-H is predicted.

Autoregressive SYM-H term

The term

$$\phi\,\mathrm{SYM{-}H}_t$$

makes the model autoregressive. It allows the current geomagnetic state to contribute directly to the prediction of future SYM-H.

This is important because SYM-H is strongly persistent over short time intervals. A storm that is already highly disturbed is generally more likely to remain disturbed in the near future than a storm currently close to quiet conditions. Including current SYM-H therefore gives the model information about the existing state of the magnetosphere before the lagged solar-wind predictors are considered.

The coefficient ϕ measures the strength of this relationship. If ϕ is large and positive, current SYM-H has a strong influence on the two-hour-ahead forecast.

The lagged Bz and solar-wind-speed terms then provide additional information about whether conditions are likely to intensify, remain stable, or recover. In this way, the final ARDL combines information about the current geomagnetic state with the recent history of the upstream solar wind.

4. Cross-Storm Validation

To evaluate whether the forecasting models generalize beyond a single geomagnetic storm, leave-one-storm-out (LOSO) cross-validation was used across the 10 selected events.

In each validation fold, one entire storm is held out as the test event, while the remaining nine storms are combined and used to train the model. The fitted model is then used to generate forecasts for the excluded storm only. This process is repeated 10 times so that every storm serves as the held-out test event once.

For example, when the 2003-11-20 storm is used as the test event, the model is fitted using the other nine storms and no observations from the 2003-11-20 event are included in the estimation stage. The same process is then repeated with each of the other storms in turn.

This procedure is important because it prevents information from the storm being evaluated from entering the fitted model. It therefore provides a stronger assessment of out-of-sample forecasting performance than fitting and evaluating the model on the same event.

The predictor histories are constructed separately within each storm before the training storms are combined. This ensures that lagged values from the end of one storm cannot incorrectly become part of the predictor history for another storm. Linear interpolation, lag construction and the future SYM-H target are therefore all performed independently within each event.

For the main 120-minute specification, the target variable is

$$\mathrm{SYM{-}H}_{t+24},$$

where 24 five-minute steps correspond to two hours. The same LOSO procedure was subsequently repeated at 60- and 180-minute forecast horizons for the sensitivity analysis in Section 5.

Forecasts are evaluated by comparing four different models.

4.1 Persistence model

The simplest benchmark is the persistence forecast:

$$\widehat{\mathrm{SYM-H}}_{t+h} = \mathrm{SYM-H}_t.$$

The persistence model assumes that the SYM-H value two hours in the future will be equal to its current value.

Persistence provides a useful benchmark because SYM-H is highly temporally persistent over short intervals. A more complicated forecasting model should ideally improve upon this simple assumption.

4.2 SYM-H-only model

The second benchmark is an autoregressive model that predicts future SYM-H using only the current SYM-H value:

$$\mathrm{SYM-H}_{t+h} = \alpha + \phi\, \mathrm{SYM-H}_t + \varepsilon_{t+h}.$$

Unlike persistence, this model does not force the future value to be exactly equal to the current value. Instead, the intercept alpha and autoregressive coefficient phi are estimated from the nine training storms.

This provides an intermediate benchmark for determining whether the solar-wind predictors add useful information beyond the current geomagnetic state alone.

4.3 Bz B-spline ARDL

The third model adds the recent history of the IMF Bz component:

$$\mathrm{SYM-H}_{t+h} = \alpha + \phi\, \mathrm{SYM-H}_t + \sum_{\ell=0}^{L} \beta_\ell Bz_{t-\ell} + \varepsilon_{t+h}.$$

The lag coefficients beta_l are represented using the B-spline basis described in the previous section.

This model therefore tests whether recent Bz conditions improve the two-hour-ahead forecast beyond the information contained in current SYM-H.

4.4 Bz + solar-wind-speed B-spline ARDL

The final model additionally includes the recent history of solar-wind flow speed:

$$\mathrm{SYM{-}H}_{t+h} = \alpha + \phi\,\mathrm{SYM{-}H}_t + \sum_{\ell=0}^{L} \beta_\ell Bz_{t-\ell} + \sum_{\ell=0}^{L} \gamma_\ell V_{t-\ell} + \varepsilon_{t+h},$$

where Vt−□V_{t-\ell} represents solar-wind speed at lag l.

This is the most complete model considered in the final validation because it combines the current geomagnetic state with recent information about both IMF Bz and solar-wind velocity.

Model evaluation

For each held-out storm, forecast accuracy is measured using root mean squared error (RMSE) and mean absolute error (MAE).

RMSE is calculated as

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2},$$

while MAE is calculated as

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|Y_i - \hat{Y}_i|.$$

Both measures are expressed in nT, with smaller values indicating more accurate forecasts.

Forecast skill relative to persistence is also calculated using

$$\mathrm{Skill} = 1 - \frac{\mathrm{RMSE}_{\mathrm{model}}}{\mathrm{RMSE}_{\mathrm{persistence}}}.$$

A positive skill value indicates that the model has a lower RMSE than persistence, while a negative value indicates that persistence performs beter.

This comparison allows the analysis to determine not only whether the ARDL models produce reasonable forecasts, but whether the additional Bz and solar-wind-speed information provides measurable predictive value beyond simpler benchmarks.

5. Results and Sensitivity Analysis

The forecasting performance of the four models was evaluated across three forecast horizons and three lag-window lengths. Forecast horizons of 60, 120 and 180 minutes were considered, corresponding to forecast_horizon = 12, 24 and 36 five-minute observations. Similarly, predictor histories of 60, 120 and 180 minutes were considered using max_lag = 12, 24 and 36.

All models were evaluated on a common set of forecast-issue times within each forecast horizon. This ensures that differences between lag-window specifications reflect changes in model performance rather than changes in the observations being evaluated.

5.1 Pooled cross-storm results

The pooled leave-one-storm-out results are summarized below.

Forecast	Lag	Model	RMSE	MAE	Skill
60	60	Persistence	19.44	8.97	0.0%
		SYM-H only	19.17	8.93	1.4%
		Bz ARDL	18.75	8.81	3.6%
		Bz + Speed ARDL	18.12	8.83	6.8%
60	120	Persistence	19.44	8.97	0.0%
		SYM-H only	19.17	8.94	1.4%
		Bz ARDL	18.71	8.66	3.7%
		Bz + Speed ARDL	18.22	8.76	6.3%
60	180	Persistence	19.44	8.97	0.0%
		SYM-H only	19.17	8.95	1.4%
		Bz ARDL	18.53	8.68	4.7%
		Bz + Speed ARDL	17.99	8.75	7.4%
120	60	Persistence	29.52	13.97	0.0%
		SYM-H only	28.83	13.98	2.3%
		Bz ARDL	28.01	14.07	5.1%
		Bz + Speed ARDL	27.30	14.29	7.5%
120	120	Persistence	29.52	13.97	0.0%
		SYM-H only	28.83	14.00	2.3%
		Bz ARDL	27.70	13.84	6.2%
		Bz + Speed ARDL	27.35	14.27	7.3%
120	180	Persistence	29.52	13.97	0.0%
		SYM-H only	28.83	14.02	2.3%
		Bz ARDL	27.85	13.90	5.7%
		Bz + Speed ARDL	27.27	14.18	7.6%
180	60	Persistence	36.67	17.93	0.0%
		SYM-H only	35.20	17.84	4.0%
		Bz ARDL	35.00	18.14	4.6%
		Bz + Speed ARDL	33.65	18.10	8.2%
180	120	Persistence	36.67	17.93	0.0%
		SYM-H only	35.20	17.87	4.0%
		Bz ARDL	34.77	17.83	5.2%
		Bz + Speed ARDL	33.60	18.15	8.4%
180	180	Persistence	36.67	17.93	0.0%
		SYM-H only	35.20	17.90	4.0%
		Bz ARDL	34.84	17.91	5.0%

Forecast	Lag	Model	RMSE	MAE	Skill
		Bz + Speed ARDL	33.74	18.25	8.0%

5.2 Effect of forecast horizon

As expected, forecasting error increased as the forecast horizon became longer. Persistence RMSE increased from approximately 19.44 nT at 60 minutes, to 29.52 nT at 120 minutes, and finally to 36.67 nT at 180 minutes.

A similar increase was observed for all fitted models. For example, the best Bz + Speed ARDL RMSE increased from approximately 17.99 nT for a 60-minute forecast, to 27.27 nT at 120 minutes, and to 33.60 nT at 180 minutes.

This indicates that predicting the geomagnetic state becomes increasingly difficult as the forecast horizon increases. However, the relative improvement of the ARDL models over persistence did not disappear at longer horizons.

In fact, the strongest relative improvement was observed for the 180-minute forecasts. The Bz + Speed model using 120 minutes of predictor history achieved an RMSE of 33.60 nT, compared with 36.67 nT for persistence, corresponding to an RMSE skill of approximately 8.4%.

5.3 Effect of lag-window length

The results do not indicate that one lag-window length is universally optimal.

For the 60-minute forecast horizon, the lowest RMSE was obtained using 180 minutes of Bz and solar-wind-speed history:

$$\text{RMSE} = 17.99 \text{ nT},$$

corresponding to approximately 7.4% improvement over persistence.

For the 120-minute forecast horizon, the Bz + Speed model again achieved its lowest RMSE with 180 minutes of predictor history:

$$\text{RMSE} = 27.27 \text{ nT},$$

although the difference from the 60- and 120-minute lag windows was small. The corresponding Bz + Speed RMSEs were 27.30, 27.35 and 27.27 nT for 60-, 120- and 180-minute lag windows respectively.

For the 180-minute forecast horizon, the best result was obtained with 120 minutes of predictor history:

$$\text{RMSE} = 33.60 \text{ nT},$$

corresponding to approximately 8.4% skill relative to persistence.

Overall, increasing the lag window from 60 to 120 or 180 minutes sometimes improved performance, but the differences were relatively small. This suggests that most of the useful predictive information may already be contained within the most recent few hours of solar-wind observations.

5.4 Contribution of Bz and solar-wind speed

Across all nine combinations of forecast horizon and lag history, the Bz + Speed ARDL achieved a lower RMSE than persistence. Its skill relative to persistence ranged from approximately 6.3% to 8.4%.

The Bz-only ARDL also consistently improved RMSE relative to persistence, although by a smaller amount. This indicates that the recent history of IMF Bz provides predictive information beyond that contained in the current SYM-H value alone.

Adding solar-wind speed generally produced a further reduction in RMSE. For example, for the 180-minute forecast using 120 minutes of history, RMSE decreased from

$$34.77 \text{ nT}$$

for the Bz-only model to

$$33.60 \text{ nT}$$

for the Bz + Speed model.

The improvement from adding speed was therefore present, but smaller than the improvement obtained by adding Bz history to the autoregressive model. Within the variables considered, this indicates that adding Bz history accounted for a larger share of the RMSE improvement than the subsequent addition of solar-wind speed, while speed still provided additional predictive information.

5.5 RMSE and MAE show slightly different behaviour

The improvement produced by the Bz + Speed model was clearer for RMSE than for MAE.

For several configurations, the Bz-only model achieved a slightly lower MAE than the Bz + Speed model. For example, for the 120-minute forecast with a 120-minute lag history:

$$\text{MAE}_{\text{Persistence}} = 13.97, \qquad \text{MAE}_{\text{Bz}} = 13.84,$$

while

$$\text{MAE}_{\text{Bz+Speed}} = 14.27.$$

Similarly, at the 180-minute horizon, the Bz + Speed model generally had a slightly higher MAE than persistence despite having substantially lower RMSE.

Since RMSE gives greater weight to large forecast errors than MAE, this pattern suggests that the addition of solar-wind speed may be particularly useful for reducing some of the larger prediction errors, while not necessarily reducing the absolute error of every individual forecast.

5.6 Overall findings

The cross-storm validation therefore produced three main findings.

First, the autoregressive component alone provides only a modest improvement over persistence. The SYM-H-only model achieved approximately 1.4% RMSE skill at 60 minutes, 2.3% at 120 minutes, and 4.0% at 180 minutes.

Second, including recent Bz observations provides additional predictive information. The Bz B-spline ARDL consistently reduced RMSE relative to both persistence and, in most cases, the SYM-H-only model.

Third, the inclusion of solar-wind speed produced the lowest RMSE in all nine lag-window/forecast-horizon combinations. The improvement was modest but consistent, with approximately 6–8% RMSE skill relative to persistence.

The best individual configurations were:

Forecast horizon	Best lag history	Bz + Speed RMSE	Persistence RMSE	Skill
60 min	180 min	17.99	19.44	7.4%
120 min	180 min	27.27	29.52	7.6%
180 min	120 min	33.60	36.67	8.4%

These results suggest that recent Bz and solar-wind-speed measurements contain useful out-of-sample information for forecasting future SYM-H across different short-term forecast horizons, although the magnitude of the improvement over persistence remains moderate.

6. Predictions and Figures

To illustrate the out-of-sample forecasting behaviour of the final model, figures were produced using the specification with a 120-minute forecast horizon and a 120-minute predictor history (forecast_horizon = 24, max_lag = 24).

A pooled predicted-versus-actual scatter plot was first constructed using all leave-one-storm-out forecasts from the ten held-out events. The 1:1 reference line represents perfect prediction. Points closer to this line indicate more accurate forecasts, while deviations indicate under- or over-prediction. Overall, the Bz + Speed ARDL predictions followed the general variation in future SYM-H, although larger errors were observed during some of the more extreme storm intervals.

A representative example is the 20 November 2003 storm. For this event, persistence produced an RMSE of 36.80 nT, while the Bz + Speed ARDL reduced the RMSE to 32.45 nT, corresponding to a skill of approximately 11.8% relative to persistence. The time-series plot therefore provides an example where inclusion of recent solar-wind information improves the forecast while still leaving noticeable forecast error.

A particularly successful case was the 31 March 2001 storm. Persistence achieved an RMSE of 45.33 nT, compared with 38.01 nT for the Bz + Speed model. This corresponds to approximately 16.1% RMSE skill, indicating that the lagged solar-wind predictors provided substantial additional information beyond the current SYM-H value for this storm.

However, the ARDL did not outperform persistence for every event. The clearest difficult case was the 23 June 2015 storm, for which persistence RMSE was 19.32 nT, while the Bz + Speed model produced an RMSE of 22.64 nT. The corresponding skill was approximately −17.2%, showing that the more complex model can perform worse than persistence for certain storm structures.

These contrasting examples demonstrate that the predictive value of recent Bz and solar-wind speed is event-dependent. While the pooled results show a consistent average improvement over persistence, the individual storm results show that the magnitude and even direction of the improvement varies across events.

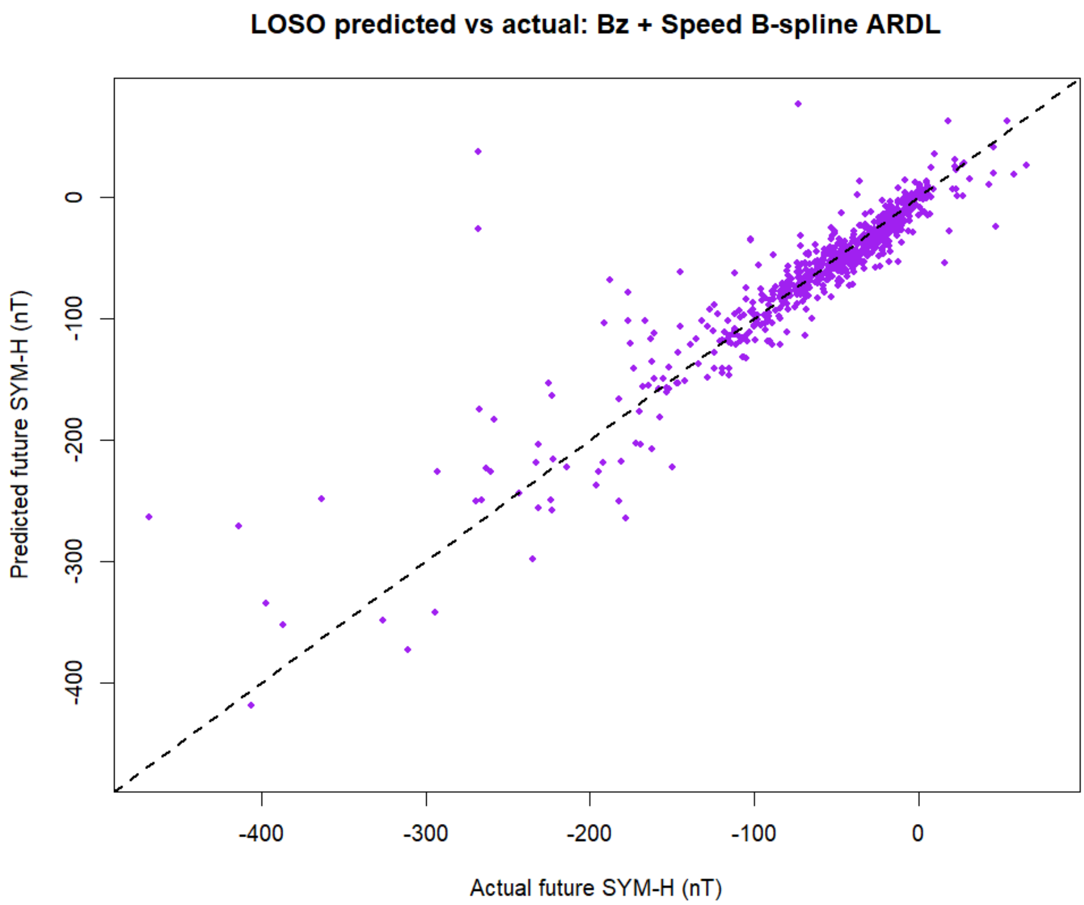

Figure 1: Pooled LOSO predicted versus actual SYM-H for the Bz + Speed B-spline ARDL using a 120-minute forecast horizon and 120-minute predictor history. The dashed 1:1 line represents perfect agreement.

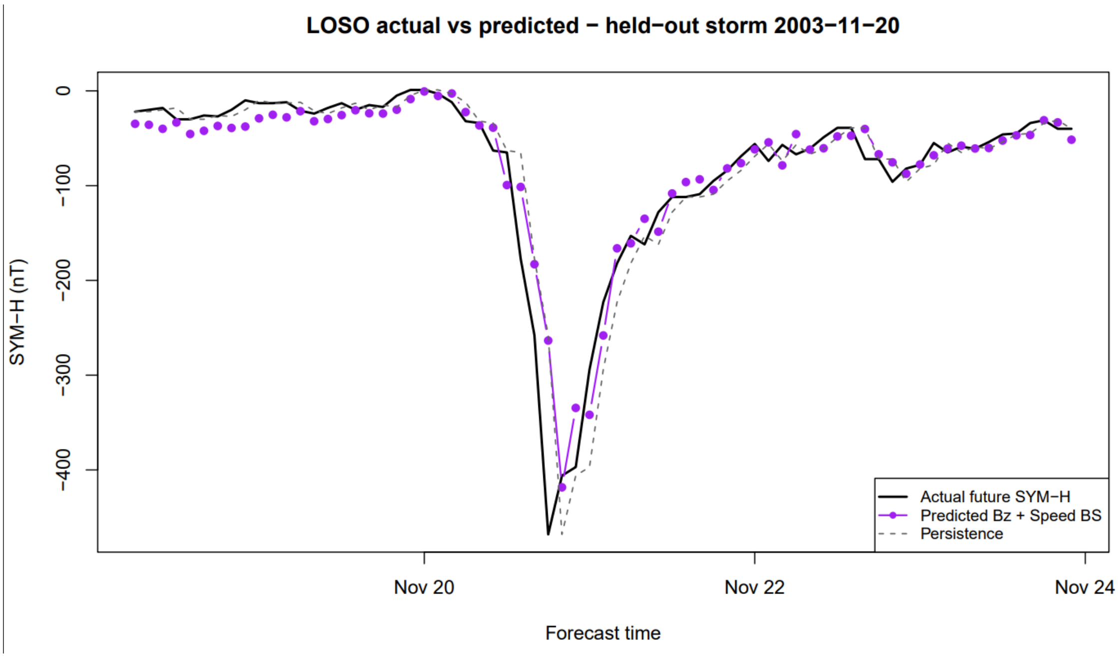

Figure 2: Actual and predicted SYM-H for the held-out 20 November 2003 storm under the 120-minute forecast and 120-minute lag specification. The plot compares the Bz + Speed ARDL with persistence.

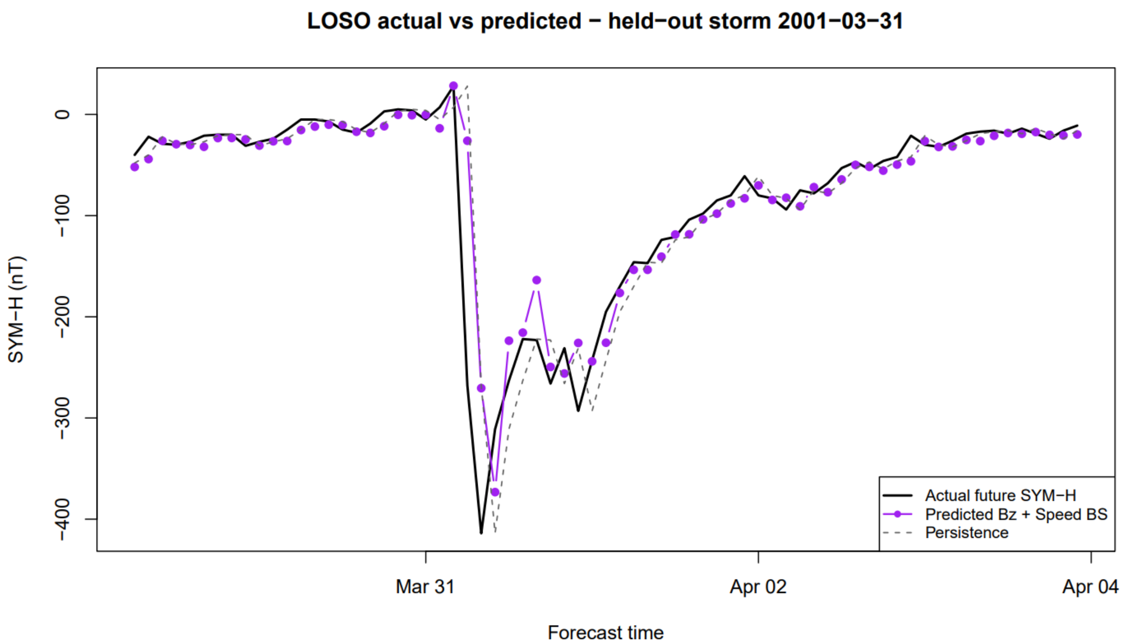

Figure 3: Actual and predicted SYM-H for the held-out 31 March 2001 storm, illustrating a case in which the Bz + Speed ARDL clearly improved on persistence.

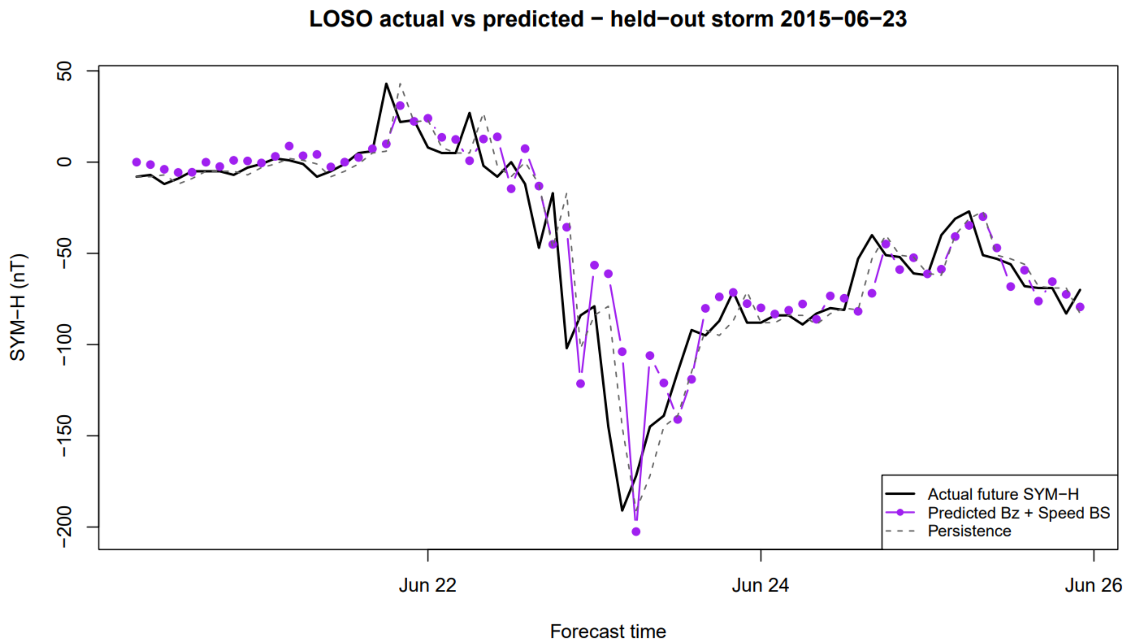

Figure 4: Actual and predicted SYM-H for the held-out 23 June 2015 storm, illustrating a difficult case in which persistence outperformed the Bz + Speed ARDL.

7. Discussion and Limitations

The purpose of this study was to investigate whether distributed lag modelling can provide a useful statistical framework for modelling and forecasting geomagnetic indices such as SYM-H. The results provide evidence that this is possible. In particular, the leave-one-storm-out validation indicates that recent solar-wind conditions contain predictive information about future SYM-H beyond that contained in the current value of SYM-H alone.

The persistence model represents a particularly important benchmark because geomagnetic indices are strongly temporally dependent over short periods. A model can appear accurate simply because the current and near-future values of SYM-H are highly correlated. For this reason, outperforming persistence is more informative than considering forecast error alone. Across all nine combinations of forecast horizon and lag-window length considered in this study, the Bz + solar-wind-speed ARDL achieved a positive RMSE skill relative to persistence, with improvements of approximately 6–8%. This suggests that the recent history of the solar wind contributes additional predictive information beyond the current geomagnetic state.

The comparison between the models also provides some indication of where this additional predictive information originates. The SYM-H-only autoregressive model produced relatively small improvements over persistence, with RMSE skill ranging from approximately 1.4% to 4.0% depending on the forecast horizon. Introducing the history of IMF Bz generally produced a larger improvement. Adding solar-wind speed resulted in a further reduction in RMSE and produced the lowest pooled RMSE for every combination of lag window and forecast horizon examined.

This behaviour is consistent with the physical motivation for the model. Geomagnetic disturbances are not determined solely by the instantaneous value of one solar-wind variable. Instead, the state of the magnetosphere reflects both its existing condition and the recent history of solar-wind forcing. A distributed lag formulation is therefore attractive because it allows the response at a future time to depend on a sequence of preceding solar-wind observations rather than only their contemporaneous values.

The sensitivity analysis also showed that there was no single lag-window length that was clearly optimal for every forecasting horizon. For a 60-minute forecast, the best Bz + Speed result used 180 minutes of predictor history, while the best 180-minute forecast used 120 minutes of history. At the 120-minute horizon, results from 60-, 120- and 180-minute lag windows were very similar. This suggests that the useful solar-wind information is distributed across the preceding few hours but that simply increasing the lag window does not necessarily improve forecasting performance.

Forecast difficulty increased substantially with the prediction horizon. The persistence RMSE increased from approximately 19.44 nT at 60 minutes to 29.52 nT at 120 minutes and 36.67 nT at 180 minutes. Nevertheless, the relative improvement of the Bz + Speed ARDL over persistence remained present at all three horizons. This is encouraging because it suggests that the predictive contribution of recent solar-wind history does not disappear when forecasts are extended from one to three hours ahead.

At the individual storm level, however, performance was heterogeneous. For example, for the 31 March 2001 storm the Bz + Speed model reduced RMSE from 45.33 nT under persistence to 38.01 nT, corresponding to approximately 16.1% skill. In contrast, for the 23 June 2015 storm the ARDL performed worse than persistence, producing negative skill of approximately −17.2%. The benefit of distributed lag modelling therefore appears to depend partly on the characteristics and evolution of the individual storm.

Another important finding is the difference between RMSE and MAE. Although the Bz + Speed model consistently improved pooled RMSE, it did not always improve MAE. For some specifications, the Bz-only model achieved a lower MAE than the Bz + Speed model, and at the longer forecast horizons the Bz + Speed MAE could even be slightly higher than persistence. Since RMSE penalizes large errors more strongly than MAE, one possible interpretation is that including solar-wind speed helps reduce some relatively large forecast errors without consistently improving every individual forecast. The benefit of speed should therefore be described as modest rather than as a uniform improvement in all measures of forecast accuracy.

Limitations

Several limitations should be considered when interpreting these results.

First, the study contains only 10 historical geomagnetic storms. Leave-one-storm-out cross-validation provides a useful test of generalization between these events, but ten storms cannot represent the full diversity of geomagnetic activity. A substantially larger sample would be required before making strong claims regarding operational forecasting performance.

Second, the storms were partly selected according to the availability and continuity of solar-wind measurements. Events containing prolonged missing intervals in Bz or flow speed were excluded, while short gaps were filled using linear interpolation. This produces a cleaner dataset for evaluating the modelling method but may introduce selection effects because storms with particularly poor data coverage are not represented.

Linear interpolation also introduces an important operational limitation. The interpolation procedure used here is retrospective, meaning that values on both sides of a missing observation can be used to estimate the missing value. In an actual real-time forecasting seting, a future observation would not yet be available. Consequently, the present results should be interpreted as an assessment of the statistical modelling framework rather than as a complete operational forecasting system.

Third, only two solar-wind predictors were included: IMF Bz and solar-wind flow speed. Geomagnetic storm development depends on additional quantities, including solar-wind density, dynamic pressure and other characteristics of the interplanetary magnetic field. Burton et al. (1975), for example, did not describe geomagnetic activity using Bz alone. Extending the distributed lag model to additional physically relevant variables may therefore improve forecasting performance.

The model also assumes an approximately additive and linear relationship between the predictor histories and future SYM-H after the B-spline transformation of the lag dimension. In reality, solar-wind–magnetosphere coupling may involve nonlinearities and interactions. For example, the physical importance of Bz can depend on solar-wind speed rather than the effects of the two quantities being completely independent. Future work could therefore examine nonlinear distributed lag models or interaction terms between Bz and speed.

Another limitation concerns the B-spline specification itself. A fixed number of spline basis functions was used to smooth the lag-response relationship. Although this reduces the instability associated with estimating a separate coefficient at every five-minute lag, alternative degrees of freedom or spline structures could produce different estimated lag-response curves. These choices could be investigated systematically in future work.

The forecasting analysis was also restricted to relatively short forecast horizons of 60, 120 and 180 minutes and predictor histories of up to three hours. The results therefore cannot establish whether the same framework would remain useful at substantially longer forecasting horizons.

Furthermore, although the events were selected using historical geomagnetic storm information, the model itself was developed for SYM-H rather than Dst. Because the two indices are closely related but differ in temporal resolution and construction, the present results should not be interpreted as direct evidence of forecasting performance for Dst. Applying the same modelling framework to Dst would require a separate analysis using its hourly observations and an appropriately defined lag structure.

Finally, the study evaluates selected storm-centered six-day intervals rather than the complete continuous solar-wind and geomagnetic record. Consequently, the models are evaluated primarily under storm-related conditions. An operational model would also need to distinguish storms from long periods of quiet or weak geomagnetic activity.

8. Conclusion

This study investigated whether distributed lag models can be applied to the modelling and short-term forecasting of geomagnetic activity. An autoregressive distributed lag framework was developed in which future SYM-H depends on its current value together with the recent history of IMF Bz and solar-wind speed. B-spline basis functions were used to represent the lag effects parsimoniously, and model generalization was evaluated using leave-one-storm-out cross-validation across ten historical geomagnetic storms.

The results provide evidence that distributed lag models can potentially be useful for modelling geomagnetic indices such as SYM-H. Across forecast horizons of 60, 120 and 180 minutes and lag histories between 60 and 180 minutes, the Bz + solar-wind-speed ARDL consistently produced a lower pooled RMSE than persistence. Its improvement relative to persistence was modest but stable, with RMSE skill of approximately 6–8%across the nine specifications examined. The Bz-only ARDL also generally outperformed the simpler benchmarks, indicating that the recent history of the interplanetary magnetic field contains predictive information beyond the current geomagnetic state.

At the same time, the results do not suggest that a distributed lag model alone provides a complete solution to geomagnetic forecasting. Performance varied considerably between storms, improvements in RMSE were not always accompanied by improvements in MAE, and the analysis was limited to ten selected events and two solar-wind predictors. The present model should therefore be interpreted as a demonstration of the potential usefulness of the distributed lag framework, rather than as an operational forecasting model.

More broadly, the results suggest that DLMs are conceptually well suited to heliophysical problems in which the response to external forcing is delayed and distributed over time. The same general framework could potentially be applied to other geomagnetic indices, including Dst, although this would require separate validation using its temporal resolution and characteristics. Future research using larger storm samples, additional solar-wind predictors, nonlinear or interacting effects, and real-time treatment of missing data could establish whether distributed lag models can provide practically useful improvements in geomagnetic storm forecasting.

> **Therefore, the main research question can be answered cautiously in the affirmative: distributed lag models appear to be a viable statistical framework for modelling and forecasting geomagnetic indices such as SYM-H, and potentially Dst, but substantially more validation is required before they could be considered operational forecasting tools.**